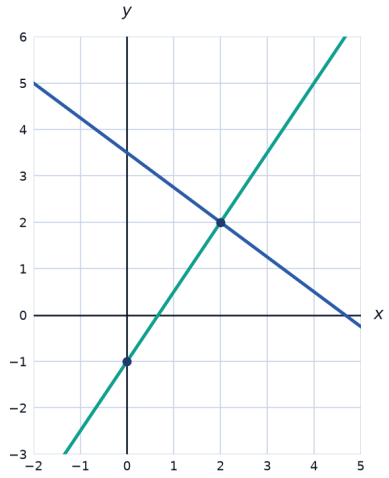
# Variables on both sides

## Core Concepts: Variables on both sides

**1** For the model $f(x) = 3x + 8$ in variables on both sides, evaluate $f(2)$.




**a.** 14

**b.** 17

**c.** 6

**d.** 18

**2** Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

**a.** $x = 1$ and $x = 5$

**b.** $x = 1$ and $x = 4$

**c.** $x = -1$ and $x = -4$

**d.** $x = 5$ and $x = 4$

**3** A data set recorded for variables on both sides consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

**a.** 21.5

**b.** 22.5

**c.** 23.5

**d.** 25.5

---

## Procedural Fluency & Algebraic Manipulation

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**4** For the model $f(x) = 3x + 8$ in variables on both sides, evaluate $f(3)$.

**a.** 17

**b.** 20

**c.** 9

**d.** 21

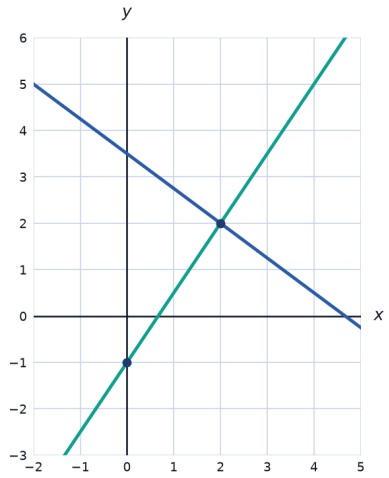
**5** Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

**a.** $x = 2$ and $x = 6$

**b.** $x = 2$ and $x = 5$

**c.** $x = -2$ and $x = -5$

**d.** $x = 7$ and $x = 10$




**6** A data set recorded for variables on both sides consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

**a.** 21.5

**b.** 22.5

**c.** 23.5

**d.** 25.5

---

## Graphical & Geometric Interpretation

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**7** For the model $f(x) = 3x + 8$ in variables on both sides, evaluate $f(4)$.

**a.** 20

**b.** 23

**c.** 12

**d.** 24

**8** Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

**a.** $x = 3$ and $x = 7$

**b.** $x = 3$ and $x = 6$

**c.** $x = -3$ and $x = -6$

**d.** $x = 9$ and $x = 18$

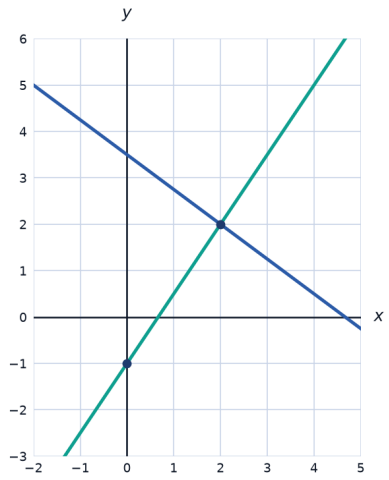
**9** A data set recorded for variables on both sides consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

**a.** 21.5

**b.** 22.5

**c.** 23.5

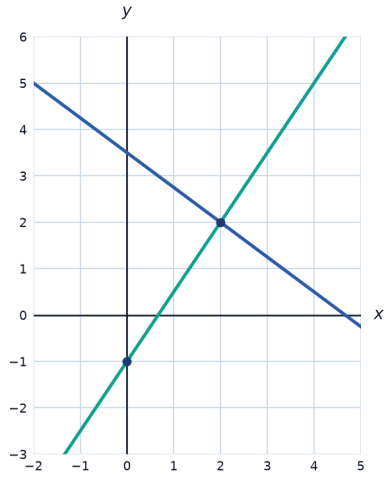
**d.** 25.5




## Exam-Style Problem Solving & Modeling

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**10** For the model $f(x) = 3x + 8$ in variables on both sides, evaluate $f(5)$.




**a.** 23

**b.** 26

**c.** 15

**d.** 27

**11** Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

**a.** $x = 4$ and $x = 8$

**b.** $x = 4$ and $x = 7$

**c.** $x = -4$ and $x = -7$

**d.** $x = 11$ and $x = 28$

**12** A data set recorded for variables on both sides consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

**a.** 21.5

**b.** 22.5

**c.** 23.5

**d.** 25.5

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